MAGSCROLL
Firmware Flasher

Mac Instructions

Flash .hex firmware over USB with automatic bootloader reset

What this does

Flashes new firmware (**.hex** file) onto the MagScroll device over USB. Automatically handles the **1200-baud bootloader reset** so you don't have to.

Requirements

- A Mac running macOS (Apple Silicon **or** Intel)
- The MagScroll plugged in via USB
- Internet access for the one-time setup step

STEP 1 — Install package manager (one-time setup)

Apple Silicon (M1 / M2 / M3 / M4) — use Homebrew

Open **Terminal** (Cmd + Space → "Terminal"). Paste this and press Enter:

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Follow the prompts, then **close and reopen Terminal** so `brew` is available.

Intel Macs — use MacPorts (Homebrew no longer supports Intel macOS)

Install MacPorts from the official site if you don't have it, then install avrdude from https://ports.macports.org/port/avrdude/

Typical install:

```
sudo port install avrdude
```

Intel paths: binary at `/opt/local/bin/avrdude`, config at `/opt/local/etc/avrdude.conf`

Note: If your package manager is already installed, skip to Step 2.

STEP 2 — Install avrdude (one-time setup)

Apple Silicon

```
brew install avrdude
```

Verify:

```
which avrdude
```

Expected: `/opt/homebrew/bin/avrdude`

Intel (MacPorts)

```
sudo port install avrdude
```

Verify:

```
which avrdude
```

Expected: `/opt/local/bin/avrdude`

The script auto-finds the config file among `/opt/homebrew/etc/avrdude.conf`, `/usr/local/etc/avrdude.conf`, and `/opt/local/etc/avrdude.conf` (MacPorts).

STEP 3 — Save the flasher script

Create a folder for the tool:

```
mkdir -p ~/MagScroll && cd ~/MagScroll
```

Create the script file:

```
nano flash_magscroll.sh
```

Paste the entire script below into the editor, then press **Ctrl + O**, Enter to save, and **Ctrl + X** to exit.

> **Important:** After the 1200-baud reset, the MagScroll may still show the original `usbmodemHID…` port. The script **ignores that HID port** and waits for a **different** bootloader port before flashing.

```
#!/bin/bash
# MagScroll Firmware Flasher for macOS
# Usage: ./flash_magscroll.sh [path/to/firmware.hex]
set -e
HEX_FILE="${1:-Firmware/MagVolume.hex}"

# --- Locate avrdude.conf ---
CONF_PATHS=(
    "/opt/homebrew/etc/avrdude.conf"
    "/usr/local/etc/avrdude.conf"
    "/opt/local/etc/avrdude.conf"
)
CONF=""
for path in "${CONF_PATHS[@]}"; do
    if [ -f "$path" ]; then
        CONF="$path"
        break
    fi
done
if [ -z "$CONF" ]; then
    echo "Could not find avrdude.conf."
    echo "   Apple Silicon: brew install avrdude"
    echo "   Intel Mac:     sudo port install avrdude"
    exit 1
fi
echo "Using config: $CONF"

# --- Verify hex file exists ---
if [ ! -f "$HEX_FILE" ]; then
    echo "Firmware file not found: $HEX_FILE"
    echo "   Place your .hex file in the Firmware/ folder,"
    echo "   or pass its path as an argument."
    exit 1
fi
echo "Firmware: $HEX_FILE"

# --- Find the MagScroll port (may be HID interface) ---
PORT=$(ls /dev/cu.usbmodem* 2>/dev/null | head -n 1)
if [ -z "$PORT" ]; then
    echo "No usbmodem device found. Is the MagScroll plugged in?"
    exit 1
fi
echo "Device found on: $PORT"
ORIGINAL_PORT="$PORT"
```

```
# --- Trigger bootloader via 1200-baud touch ---
echo "Triggering bootloader..."
stty -f "$PORT" 1200 || true

# --- Wait for a DIFFERENT bootloader port (not the HID port) ---
echo "Waiting for bootloader..."
BOOTLOADER_PORT=""
for i in $(seq 1 20); do
    sleep 0.5
    for CANDIDATE in /dev/cu.usbmodem*; do
        [ -e "$CANDIDATE" ] || continue
        [ -c "$CANDIDATE" ] || continue
        # Skip the original HID / app port — flash on the new one only
        if [ "$CANDIDATE" = "$ORIGINAL_PORT" ]; then
            continue
        fi
        BOOTLOADER_PORT="$CANDIDATE"
        break
    done
    if [ -n "$BOOTLOADER_PORT" ]; then
        sleep 0.5
        break
    fi
done
if [ -z "$BOOTLOADER_PORT" ]; then
    echo "Bootloader port did not appear (or only HID port is present)."
    echo "   Try unplugging and replugging the MagScroll,"
    echo "   then run this script again."
    exit 1
fi
echo "Bootloader active on: $BOOTLOADER_PORT"

# --- Flash ---
echo "Flashing firmware..."
avrdude -C "$CONF" \
        -p atmega32u4 \
        -c avr109 \
        -P "$BOOTLOADER_PORT" \
        -b 57600 \
        -U flash:w:"$HEX_FILE":i

echo ""
echo "Firmware flashed successfully!"
echo "   You can now unplug the MagScroll and test it."
```

Make it executable:

```
chmod +x flash_magscroll.sh
```

STEP 4 — Add firmware files

Create a **Firmware** folder inside `~/MagScroll` and place your **.hex** file(s) there:

```
mkdir -p ~/MagScroll/Firmware
```

Copy your `MagVolume.hex` (or any other .hex files) into that folder. You can drag-and-drop in Finder, or use `cp` in Terminal.

STEP 5 — Flash the MagScroll

1. Plug the MagScroll into your Mac via USB.
2. In Terminal, from the `~/MagScroll` folder, run:

```
./flash_magscroll.sh
```

That uses the default `Firmware/MagVolume.hex`.

To flash a **different** hex file:

```
./flash_magscroll.sh Firmware/OtherFirmware.hex
```

Or an absolute path:

```
./flash_magscroll.sh ~/Downloads/MyCustom.hex
```

Example successful output (note different ports):

```
Using config: /opt/local/etc/avrdude.conf
Firmware: Firmware/MagVolume.hex
Device found on: /dev/cu.usbmodemHIDJB1
Triggering bootloader...
Waiting for bootloader...
Bootloader active on: /dev/cu.usbmodem1463101
Flashing firmware...
...
Firmware flashed successfully!
```

The first port is often the HID/app interface; flashing must use the **second, different** bootloader port.

Troubleshooting

"No usbmodem device found"

- Make sure the MagScroll is plugged in.
- Try a different USB cable (some cables are power-only).
- Run: `ls /dev/cu.usbmodem*` — if nothing shows, macOS isn't seeing the device.

"Bootloader port did not appear"

- Unplug the MagScroll, wait 5 seconds, plug it back in, run the script again.
- Confirm a **new** port appears after the 1200-baud trigger (not only the HID port).
- If it still fails, try double-tapping the MagScroll's reset button right after running the script (if it has one).

"Could not find avrdude.conf"

- Apple Silicon: `brew install avrdude` → config usually `/opt/homebrew/etc/avrdude.conf`
- Intel: `sudo port install avrdude` → config `/opt/local/etc/avrdude.conf`
- Port info: `https://ports.macports.org/port/avrdude/`

"avrdude: command not found"

- Apple Silicon: ensure Homebrew is in PATH, then `brew install avrdude`
- Intel: ensure MacPorts is in PATH (`/opt/local/bin`), then `sudo port install avrdude`

"Permission denied" when running the script

- Run: `chmod +x flash_magscroll.sh`

Optional — Double-click launcher

If you prefer double-clicking over Terminal, create a file called **Flash MagScroll.command** next to the script:

```
#!/bin/bash
cd "$(dirname "$0")"
./flash_magscroll.sh
echo ""
echo "Press any key to close..."
read -n 1
```

Make it executable:

```
chmod +x "Flash MagScroll.command"
```

Now you can double-click **Flash MagScroll.command** in Finder and it will run in Terminal automatically.

Summary — files to keep together

1. This instruction document
2. The `flash_magscroll.sh` script
3. The `Firmware/` folder containing your .hex file(s)

> Keep all three in the same **~/MagScroll** folder and the tool works out of the box.

— End of document —